

Introduction To NumPy: Arrays & Operations



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3 Feb, 2026 At 10:00 PM

Pre-Reads

Recommended

Resource



Associated Lectures (1)



Description



Discussions

Session 3.1: Introduction to NumPy: Arrays & Operations

Pre-Read Material

Program: AIM101 — Artificial Intelligence & Machine Learning Foundation **Institution:** Vishlesan i-Hub IIT Patna x Masai School

What You Will Learn Today

In this session, you will learn the fundamentals of NumPy—the essential library for numerical computing in Python. By the end of class, you will be able to:

- Explain why NumPy is faster than Python lists for numerical operations
- Create NumPy arrays using multiple methods
- Inspect array attributes (shape, dtype, ndim, size)
- Perform element-wise arithmetic and comparison operations
- Apply basic broadcasting rules
- Create simple vector visualizations

Business Story: The \$881 Million Lesson

The Zillow Algorithm Disaster (2021)

Zillow, one of the largest real estate companies in the United States, launched "Zillow Offers" — an ambitious program to use algorithms to buy and sell homes automatically. The company believed it could use data and numerical modeling to predict home prices better than anyone else.

What went wrong?

The algorithm relied on processing massive amounts of numerical data: home prices, market trends, neighborhood statistics. But the team made critical errors in how they processed and modeled this numerical data. The algorithm consistently overpaid for homes, unable to adapt quickly enough to market changes.

The result?

- November 2021: Zillow announced it would shut down Zillow Offers
- Loss: \$881 million in Q3 2021 alone
- Impact: 2,000 employees lost their jobs
- Stock price: Dropped 25%

The lesson for you:

Understanding your numerical tools isn't just academic—it's essential. The data scientists at Zillow were smart people using sophisticated tools, but mistakes in numerical processing cost the company nearly a billion dollars. As you learn NumPy, remember: every operation you perform on data matters.

Setup Requirements

Before Class, Make Sure You Have:

Python 3.12.x installed (verify with `python --version`)

NumPy installed:

```
pip install numpy
```

Matplotlib installed (for visualization):

```
pip install matplotlib
```

Jupyter Notebook ready:

```
pip install jupyter
```

Verify Your Setup

Open a Python interpreter or Jupyter notebook and run:

```
import numpy as np
import matplotlib.pyplot as plt

print(f"NumPy version: {np.__version__}")
print("Setup complete!")
```

If you see the version number and "Setup complete!" without errors, you're ready!

Concepts to Preview

What is NumPy?

NumPy (Numerical Python) is the fundamental library for scientific computing in Python. It provides:

- **High-performance multidimensional arrays** — containers for storing numerical data
- **Mathematical functions** — operations that work on entire arrays at once
- **Linear algebra tools** — matrix operations essential for machine learning

Why Not Just Use Python Lists?

You've already learned about Python lists. They're flexible and can hold any type of data. But for numerical computing, they have a major drawback: **they're slow**.

Think about this scenario:

- You have 1 million temperature readings
- You need to convert all of them from Celsius to Fahrenheit

With a Python list, you'd need a loop that processes each element one at a time. With NumPy, you write one line of code, and ALL million conversions happen simultaneously (well, almost!).

Arrays vs Lists: Key Differences

Feature	Python List	NumPy Array
Element types	Can be mixed	Must be same type
Memory	Scattered	Contiguous
Speed	Slow for math	50-100x faster
Operations	Concatenation/repetition	Element-wise math

What is an Array?

An array is simply a container for storing multiple values of the same type. Think of it like:

- **1D array (vector):** A single row of values [1, 2, 3, 4, 5]
- **2D array (matrix):** A table with rows and columns
- **3D array (tensor):** A cube of values (like a stack of tables)

Key Vocabulary

Before class, familiarize yourself with these terms:

Term	Definition
Array	A collection of values, all of the same type, arranged in a grid
Vector	A 1-dimensional array (single row or column of values)
Matrix	A 2-dimensional array (table with rows and columns)
Tensor	An n-dimensional array (generalization of vectors and matrices)
Element-wise	An operation applied to each element individually
Vectorization	Performing operations on entire arrays without explicit loops
Broadcasting	NumPy's ability to work with arrays of different shapes
dtype	The data type of elements in an array (e.g., int64, float64)
Shape	The dimensions of an array (e.g., (3, 4) means 3 rows, 4 columns)

Things to Ponder

Before class, think about these questions:

Speed matters: If processing 1 million numbers takes 100ms with a list and 1ms with NumPy, how much time would you save processing 1 billion numbers?

Trade-offs: Lists can hold mixed types (integers, strings, objects). NumPy arrays cannot. When might this be a problem?

Memory: Why do you think storing values in contiguous (side-by-side) memory is faster than scattered memory?

Real applications: Where do you think extremely fast numerical processing is essential? (Hint: think about finance, science, gaming, AI)

What's Coming After This Session

This session introduces NumPy fundamentals. In the next session (3.2), you'll learn:

- Advanced indexing techniques (fancy indexing, boolean masking)
- Reshaping arrays
- Universal functions (ufuncs)
- Memory views and performance optimization

By the end of Week 3, you'll be comfortable with NumPy and ready to tackle Pandas in Week 4.

Optional Deep Dive

If you want to explore before class:

NumPy Official Documentation: <https://numpy.org/doc/stable/>

NumPy Quickstart Tutorial: <https://numpy.org/doc/stable/user/quickstart.html>

Why NumPy is Fast (Technical):

- NumPy uses BLAS (Basic Linear Algebra Subprograms)
 - Operations are written in C and Fortran
 - Memory layout is optimized for CPU cache
-

Pre-Class Checklist

Before attending class, confirm:

- ☐ Python 3.12.x is installed
 - ☐ NumPy is installed and imports without error
 - ☐ Matplotlib is installed
 - ☐ Jupyter Notebook is working
 - ☐ I understand what an array is (conceptually)
 - ☐ I've read the Zillow case study
 - ☐ I know the key vocabulary terms
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